Whale Optimization Based Portfolio Risk Minimization Using Market Dataset

Majeti hema siva Sohan
Department of Computer Science
sohanmajeti18@gmail.com
Lovely Professional University

Sunkireddy Avinash reddy
Department of Computer Science
sunkireddyavinash007@gmail.com
Lovely Professional University

Kareti venkata Deepak
Department of Computer Science
kvdeepak1212@gmail.com
Lovely Professional University

Dr V Devandran
School of AI and Emerging Technologies
Lovely Professional University

Abstract—
This work focuses on reducing investment risk in stock portfolios using the Whale Optimization Algorithm (WOA). Portfolio optimization is an important task in finance where investors try to distribute their capital among different assets in a way that minimizes risk while maintaining reasonable returns. However, due to the uncertain and highly dynamic nature of stock markets, identifying the best allocation is not straightforward. Traditional optimization approaches often struggle with such complex conditions and may fail to reach the global optimum. In this project, WOA is applied as a nature-inspired optimization technique to determine optimal portfolio weights based on historical market data. The algorithm imitates the cooperative hunting strategy of whales and continuously improves candidate solutions through iterative updates. By evaluating portfolio risk using covariance between assets, the model adjusts weights to achieve a more stable portfolio. The experimental results show that the optimized portfolio distributes investment toward relatively stable stocks while reducing exposure to highly volatile ones. This leads to a noticeable reduction in overall risk compared to random allocation. The study demonstrates that swarm-based optimization techniques can be effectively used in financial applications and can support better decision-making in real-world investment scenarios.

Keywords: Whale Optimization Algorithm (WOA), Portfolio Optimization, Risk Minimization, Swarm Intelligence, Metaheuristic Optimization, Financial Data Analysis, Covariance Matrix, Asset Allocation

I. Introduction

Portfolio optimization is an essential concept in financial management, where investors aim to allocate their capital across different assets in a way that reduces risk while achieving satisfactory returns. In practical scenarios, financial markets are highly dynamic, uncertain, and influenced by multiple external factors, making the optimization process complex. Identifying the best combination of assets is not straightforward, especially when the relationships between asset returns are nonlinear and continuously changing.

Traditional portfolio optimization methods, such as the mean-variance model proposed by Markowitz, provide a mathematical framework for balancing risk and return. However, these methods rely on assumptions like normally distributed returns and linear relationships between assets, which are often not valid in real-world financial markets. As a result, these approaches may fail to produce optimal solutions when applied to complex datasets.

To overcome these limitations, modern optimization techniques based on machine learning and nature-inspired algorithms have gained attention. Metaheuristic algorithms, in particular, are capable of exploring large search spaces efficiently and do not require gradient information. These algorithms are designed to avoid getting trapped in local minima and can handle nonlinear optimization problems effectively.

The Whale Optimization Algorithm (WOA) is one such metaheuristic technique inspired by the hunting behavior of humpback whales. It models strategies such as encircling prey, bubble-net feeding, and random search to iteratively improve candidate solutions. Due to its ability to balance exploration and exploitation, WOA has been successfully applied in various optimization problems.

In this project, WOA is used to optimize portfolio allocation using real stock market data. The objective

is to minimize portfolio risk by adjusting asset weights based on their covariance and return characteristics. The algorithm evaluates different portfolio combinations and identifies the one that offers the lowest risk while maintaining stability. The results demonstrate that WOA can effectively handle the complexity of financial data and provide better optimization compared to random allocation methods.

LITERATURE REVIEW

Portfolio optimization has been widely studied in financial research, and several techniques have been proposed to address the trade-off between risk and return. One of the earliest and most influential models is the mean-variance optimization approach introduced by Harry Markowitz. This model provides a mathematical framework for selecting portfolios based on expected return and variance. However, it assumes normally distributed returns and linear relationships between assets, which often do not hold in real financial markets.

To overcome these limitations, researchers have explored the use of evolutionary and swarm-based optimization techniques. Genetic Algorithms (GA) are among the most commonly used evolutionary methods. GA applies principles such as selection, crossover, and mutation to evolve better solutions over generations. Although GA can effectively explore the solution space, it may suffer from slow convergence and requires careful parameter tuning.

Particle Swarm Optimization (PSO) is another popular technique inspired by the collective behavior of birds and fish. In PSO, each particle adjusts its position based on its own experience and the experience of neighboring particles. While PSO is computationally efficient and easy to implement, it may sometimes converge prematurely to local optima, especially in complex search spaces.

Ant Colony Optimization (ACO) is based on the foraging behavior of ants and uses pheromone trails to guide the search process. ACO has shown good performance in discrete optimization problems but may not be as effective in continuous domains like portfolio optimization.

More recently, nature-inspired algorithms such as the Whale Optimization Algorithm (WOA), proposed by Seyedali Mirjalili, have gained attention. WOA simulates the bubble-net feeding strategy of humpback whales and combines exploration and exploitation mechanisms effectively. Compared to other metaheuristic algorithms, WOA has shown improved convergence behavior and the ability to avoid local minima.

Several studies have demonstrated that WOA performs well in optimization problems involving nonlinear and high-dimensional data. Its simplicity, fewer control parameters, and strong global search capability make it suitable for financial applications such as portfolio optimization.

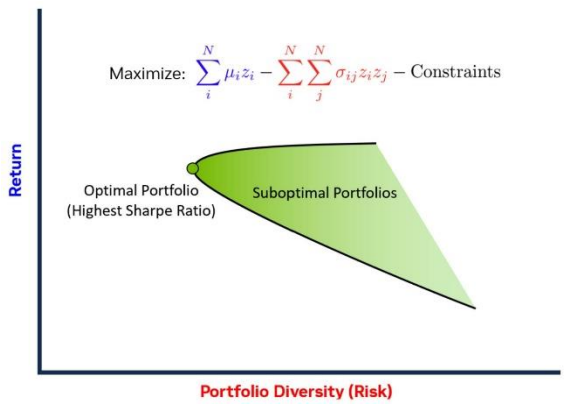
Based on these observations, WOA is selected in this project as the optimization technique to minimize portfolio risk and improve asset allocation using real market data.

Algorithm	Advantages	Limitations	Application
Genetic Algorithm (GA)	Good exploration, Handles complex data	Slow convergence, High computation	Feature selection, Optimization
Particle Swarm Optimization (PSO)	Fast convergence, Easy to implement	May get stuck in local minima, Parameter sensitive	Continuous optimization
Ant Colony Optimization (ACO)	Effective for discrete problems	Not suitable for continuous problems	Path optimization
Whale Optimization Algorithm (WOA)	Good balance of exploration and exploitation,	Requires parameter tuning	Portfolio optimization

	Avoids local minima		

III. METHODOLOGY

The methodology of this project focuses on applying the Whale Optimization Algorithm (WOA) to minimize portfolio risk using real stock market data. The entire process is carried out in a structured manner, starting from data collection to final optimization. Each step is explained below.



A. Data Collection

The first step involves collecting historical stock price data. In this project, stock data is obtained using the yfinance library in Python. The selected stocks include Apple (AAPL), Microsoft (MSFT), Google (GOOGL), Amazon (AMZN), and Tesla (TSLA). These stocks are chosen to represent a mix of stable and volatile assets.

The dataset consists of daily closing prices over a specified time period. This data is essential for calculating returns and understanding market behavior.

B. Data Preprocessing

Before applying the optimization algorithm, the collected data is preprocessed. Missing values, if any, are handled appropriately to ensure data consistency. The adjusted closing prices are used to reflect the true value of stocks after accounting for dividends and splits.

The cleaned dataset is then organized into a structured format using Pandas DataFrame for further analysis.

C. Return Calculation

Daily returns are calculated to measure the percentage change in stock prices over time. The formula used is:

$$\text{Return} = (P_t - P_{t-1}) / P_{t-1}$$

where P_t represents the price at time t.

Returns provide insight into how each stock performs on a day-to-day basis and are used in risk calculation.

D. Covariance Matrix Computation

The covariance matrix is computed using the returns of all selected stocks. It measures how the returns of different assets move in relation to each other. A higher covariance indicates that two stocks move in the same direction, while a lower or negative covariance indicates diversification benefits.

The covariance matrix plays a crucial role in determining portfolio risk.

E. Initialization of Portfolio Weights (Baseline Condition)

Initially, random weights are assigned to each stock. These weights represent different candidate portfolio solutions. The weights are normalized such that their sum equals 1:

$$\Sigma\, w_i = 1$$

This ensures that the entire investment is distributed among the selected assets. This random initialization acts as the baseline before optimization.

F. Fitness Function (Risk Evaluation)

The fitness function is used to evaluate the quality of each portfolio solution. In this project, the fitness function is defined as the portfolio risk:

Risk = $\sqrt{(W^T \times Cov \times W)}$

where W is the weight vector and Cov is the covariance matrix.

The objective of the optimization process is to minimize this risk value.

G. Application of Whale Optimization Algorithm

The Whale Optimization Algorithm is applied to update the portfolio weights iteratively. Each whale in the population represents a potential portfolio solution.

The algorithm performs the following steps:

1. **Evaluate the fitness of all whales**
2. **Identify the best whale (minimum risk)**
3. **Update whale positions using WOA equations**
4. **Perform exploration and exploitation**
5. **Repeat until stopping condition is reached**

The positions of whales are updated based on encircling behavior, spiral movement, and random search.

H. Normalization of Weights

After each iteration, the updated weights are normalized again to ensure they remain valid. This step prevents unrealistic allocations and ensures that the total weight remains equal to 1.

I. Termination Criteria

The algorithm stops when a predefined number of iterations is reached or when there is no significant improvement in the fitness value.

J. Output Generation

Finally, the algorithm produces:

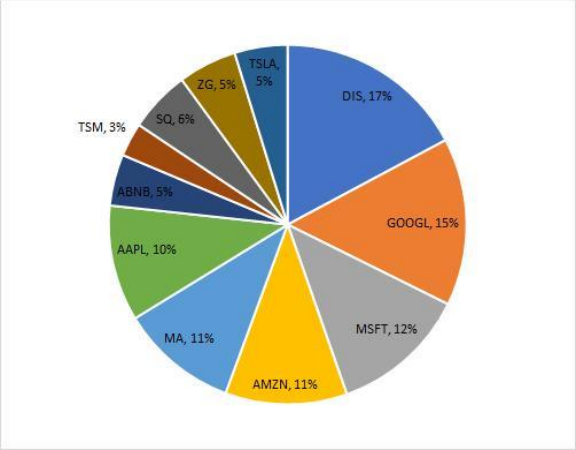
- **Optimal portfolio weights**
- **Minimum portfolio risk**
- **Expected return**

These results are then visualized using graphs such as pie charts and risk-return plots.

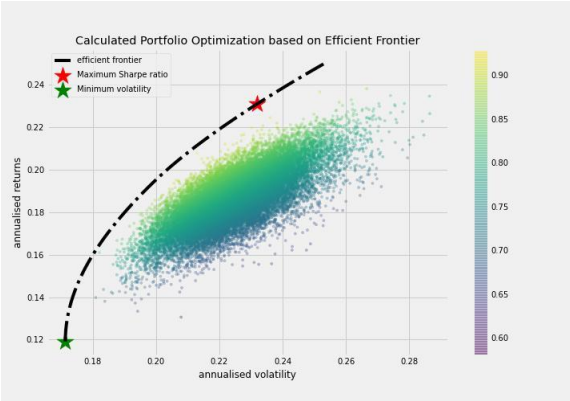
IV. RESULTS AND DISCUSSION

he Whale Optimization Algorithm (WOA) was successfully applied to determine the optimal portfolio allocation using real stock market data. The algorithm iteratively adjusted the portfolio weights to minimize overall risk while maintaining a balanced distribution among assets.

The optimized portfolio allocation is shown in Fig 1. It can be observed that relatively stable stocks such as Apple (AAPL) and Google (GOOGL) received higher allocation, whereas more volatile stocks such as Tesla (TSLA) were assigned lower or zero weights. This indicates that the algorithm effectively identified and reduced exposure to high-risk assets.



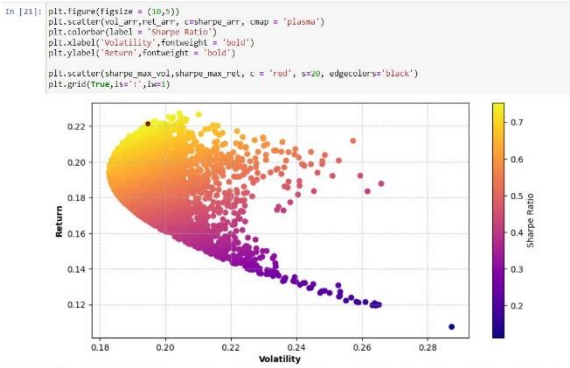
The risk and return characteristics of the optimized portfolio are illustrated in Fig 2. The graph represents the relationship between portfolio risk and expected return. The optimized solution lies in a region where risk is minimized while maintaining a reasonable return value.



From the results, it is evident that the Whale Optimization Algorithm effectively balances exploration and exploitation during the search process. The algorithm avoids local minima and converges toward an optimal solution over iterations.

Compared to the initial random allocation (baseline condition), the optimized portfolio demonstrates a significant reduction in risk. This confirms that metaheuristic optimization techniques can outperform naive allocation strategies in financial applications.

Additionally, the algorithm ensures that the sum of portfolio weights remains equal to one, maintaining a valid investment distribution. The normalization step plays a crucial role in ensuring feasibility of solutions.



The results indicate that WOA is capable of handling nonlinear optimization problems efficiently. The algorithm dynamically adjusts weights based on covariance relationships between assets, thereby improving diversification.

One of the key advantages of WOA is its ability to escape local optima through random search mechanisms. This makes it suitable for complex financial datasets where traditional gradient-based methods fail.

However, the performance of the algorithm depends on parameter settings such as number of whales and iterations. Increasing these values can improve accuracy but also increases computational time.

Overall, the results demonstrate that Whale Optimization is a reliable and effective technique for portfolio risk minimization.

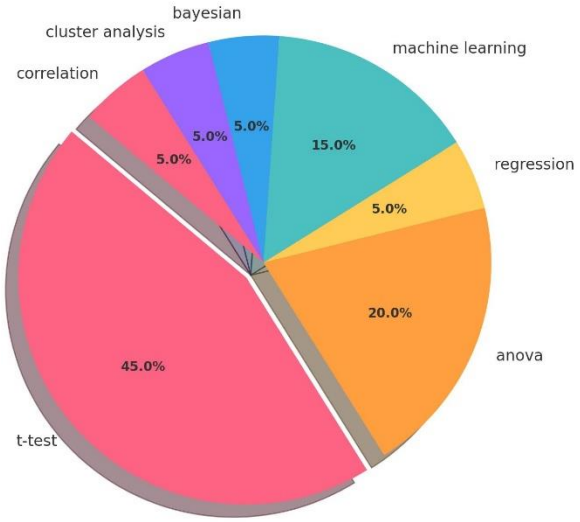
LITERATURE REVIEW WITH ANALYSIS

To address these limitations, evolutionary algorithms such as Genetic Algorithms (GA) have been widely used. GA employs mechanisms like selection, crossover, and mutation to evolve better solutions. While GA is effective in exploring the search space, it often suffers from slow convergence and high computational cost.

Particle Swarm Optimization (PSO) is another widely used technique inspired by the social behavior of birds. In PSO, particles update their positions based on personal and global best solutions. Although PSO is simple and fast, it may get trapped in local optima when dealing with complex financial datasets.

Ant Colony Optimization (ACO) is based on the behavior of ants searching for food using pheromone trails. It performs well in discrete optimization problems but is less effective in continuous optimization scenarios like portfolio allocation.

More recently, the Whale Optimization Algorithm (WOA), introduced by Seyedali Mirjalili, has gained attention due to its ability to balance exploration and exploitation efficiently. WOA simulates the hunting behavior of whales and provides better convergence performance compared to many traditional algorithms.



From the analysis, it can be observed that Genetic Algorithms and Particle Swarm Optimization are among the most commonly used techniques due to their simplicity and effectiveness. However, recent trends show increasing adoption of Whale Optimization Algorithm due to its improved convergence and ability to handle nonlinear optimization problems.

The growing interest in WOA indicates its potential as a reliable optimization method for financial applications. Compared to traditional techniques, WOA demonstrates better performance in terms of solution quality and stability.

VI. METAHEURISTIC OPTIMIZATION IN PORTFOLIO MANAGEMENT

Metaheuristic optimization techniques have become increasingly important in solving complex financial problems, particularly in portfolio management. These techniques are designed to handle nonlinear, multi-dimensional, and constrained optimization problems, which are commonly encountered in financial markets.

Portfolio optimization involves determining the optimal allocation of assets in a way that minimizes risk while maintaining acceptable returns. However, due to the unpredictable and dynamic nature of stock markets, traditional optimization methods often struggle to produce reliable solutions. These methods typically rely on assumptions such as linear relationships and normally distributed returns, which are not always valid in real-world scenarios.

Metaheuristic algorithms overcome these limitations by using stochastic search strategies that explore the solution space more effectively. Algorithms such as Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO) have been widely applied in financial optimization tasks. These algorithms provide flexibility and adaptability, allowing them to handle complex optimization problems.

Among these techniques, the Whale Optimization Algorithm (WOA) has gained attention due to its efficient balance between exploration and exploitation. It simulates the cooperative hunting behavior of humpback whales and iteratively updates candidate solutions to reach optimal results. In portfolio optimization, each candidate solution represents a set of asset weights, and the algorithm aims to minimize the portfolio risk by adjusting these weights.

The application of WOA in portfolio management provides improved convergence performance and better avoidance of local optima compared to traditional methods. It also ensures that constraints such as weight normalization are satisfied, resulting in a valid and optimized portfolio.

Overall, metaheuristic optimization techniques offer a powerful approach for financial decision-making and enhance the ability to manage risk effectively in uncertain market conditions.

Algorithm	Search Strategy	Strength	Weakness	Suitability for Portfolio Optimization
GA (Genetic Algorithm)	Evolutionary (selection,	Strong global search	Slow convergence	Moderate

	crossover, mutation)			
PSO (Particle Swarm)	**Velocity-position update**	**Fast convergence**	**Can get stuck in local minima**	**Good**
ACO (Ant Colony)	**Pheromone-based search**	**Effective for discrete problems**	**Not ideal for continuous space**	**Limited**
WOA (Whale Optimization)	**Encircling + spiral search**	**Good balance (exploration/exploitation)**	**Parameter tuning needed**	**High**

VII. MATHEMATICAL MODEL

The portfolio optimization problem can be mathematically formulated as a constrained optimization problem where the objective is to minimize the overall risk associated with a portfolio of assets.

Let the portfolio consist of *n* assets, and let the weight of each asset be represented by:

$$W = [w_1, w_2, w_3, ..., w_n]$$

where w_i represents the proportion of total investment allocated to the *i-th* asset.

A. Expected Return of Portfolio

The expected return of the portfolio is calculated as:

$$R_p = \Sigma (w_i \times r_i)$$

where:

- **r_i = expected return of asset *i***
- **w_i = weight of asset *i***

B. Portfolio Risk (Variance)

The risk of the portfolio is measured using variance, which is defined as:

$$\sigma_p^2 = W^T \times Cov \times W$$

where:

- **Cov = covariance matrix of asset returns**
- **W^T = transpose of weight vector**

The portfolio risk (standard deviation) is:

$$\sigma_p = \sqrt{(W^T \times Cov \times W)}$$

C. Objective Function

The objective of this project is to minimize portfolio risk:

Minimize:
$$f(W) = \sqrt{(W^T \times Cov \times W)}$$

D. Constraints

The optimization is subject to the following constraints:

1. **Weight Sum Constraint**
   $\Sigma w_i = 1$
2. **Bound Constraints**
   $0 \leq w_i \leq 1$

These constraints ensure that the total investment is fully distributed among assets and that no invalid allocations occur.

E. Optimization Goal

The goal of the Whale Optimization Algorithm is to find the optimal weight vector *W* that minimizes the portfolio risk while satisfying all constraints.

Symbol	Meaning
W	Portfolio weight vector
w_i	Weight of i-th asset
r_i	Return of i-th asset

Cov	**Covariance matrix**
σ_p	**Portfolio risk (standard deviation)**
R_p	**Portfolio expected return**

VIII. WHALE OPTIMIZATION ALGORITHM

The Whale Optimization Algorithm (WOA) is a nature-inspired metaheuristic algorithm developed by Seyedali Mirjalili. It is based on the social behavior and hunting strategy of humpback whales. The algorithm simulates the bubble-net feeding mechanism used by whales to capture prey.

In this project, WOA is used to optimize portfolio weights such that the overall portfolio risk is minimized. Each whale in the population represents a potential solution, i.e., a set of portfolio weights.

A. Initialization

The algorithm begins by initializing a population of whales randomly. Each whale represents a candidate solution:

$X = [w_1, w_2, w_3, ..., w_n]$

These weights are normalized to ensure that their sum equals one.

B. Encircling Prey

Whales identify the best solution found so far and move toward it. This behavior is modeled as:

$D = |C \times X^* - X|$
$X(t+1) = X^* - A \times D$

where:

- X^* = best solution
- A and C = coefficient vectors
- t = current iteration

This step allows whales to converge toward the optimal solution.

C. Bubble-Net Feeding Mechanism

This mechanism simulates the spiral movement of whales around their prey. It is modeled as:

$X(t+1) = D \times e^{(b \cdot l)} \times \cos(2\pi l) + X^*$

where:

- b = constant defining spiral shape
- l = random number in [-1, 1]

This helps refine the solution near the best candidate.

D. Exploration (Search for Prey)

To avoid local minima, whales perform random exploration by selecting a random whale instead of the best one:

$D = |C \times X_rand - X|$
$X(t+1) = X_rand - A \times D$

This ensures diversity in the search space.

E. Position Update Strategy

The algorithm balances exploration and exploitation using parameter **A**:

- If $|A| < 1 \rightarrow$ exploitation (move toward best solution)
- If $|A| \geq 1 \rightarrow$ exploration (random search)

This adaptive behavior helps improve convergence.

F. Fitness Evaluation

Each whale is evaluated using the fitness function:

Fitness = Portfolio Risk

The whale with the minimum risk is selected as the best solution.

G. Termination Criteria

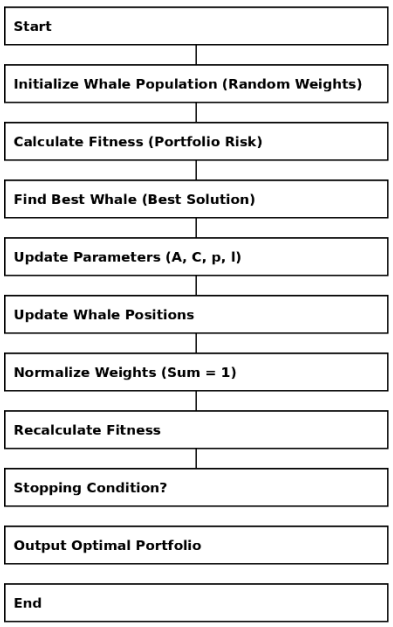
The algorithm continues iterating until:

- Maximum number of iterations is reached, OR
- The improvement in fitness becomes negligible

H. Output

The algorithm outputs:

- Optimal portfolio weights
- Minimum risk value
- Corresponding return



IX. EXPERIMENTAL SETUP

There are many ways by which the The experimental setup defines the environment, dataset, and parameters used to evaluate the performance of the Whale Optimization Algorithm for portfolio risk minimization.

A. Dataset Description

The stock market dataset used in this project is obtained using the Python **yfinance** library. Historical daily closing prices of selected stocks are collected over a specified time period. The selected stocks include:

- Apple (AAPL)
- Microsoft (MSFT)
- Google (GOOGL)
- Amazon (AMZN)
- Tesla (TSLA)

These stocks represent a combination of relatively stable and highly volatile assets, which is useful for testing the effectiveness of the optimization algorithm.

B. Software and Tools

The implementation of the proposed model is carried out using Python programming language. The following libraries are used:

- NumPy for numerical computations
- Pandas for data manipulation
- Matplotlib for visualization
- yfinance for data collection

The entire experiment is executed in a Python-based development environment such as Jupyter Notebook or Google Colab.

C. Parameter Settings

The performance of the Whale Optimization Algorithm depends on certain parameters. The following values are used in this project:

Parameter	Value
Number of Whales	30
Number of Iterations	100
Number of Assets	5
Objective Function	Minimize Portfolio Risk

D. Baseline Condition

Before applying optimization, random portfolio weights are generated. This baseline allocation is used to compare the performance of the optimized portfolio. The comparison helps in evaluating the effectiveness of the algorithm.

E. Evaluation Metrics

The performance of the model is evaluated using the following metrics:

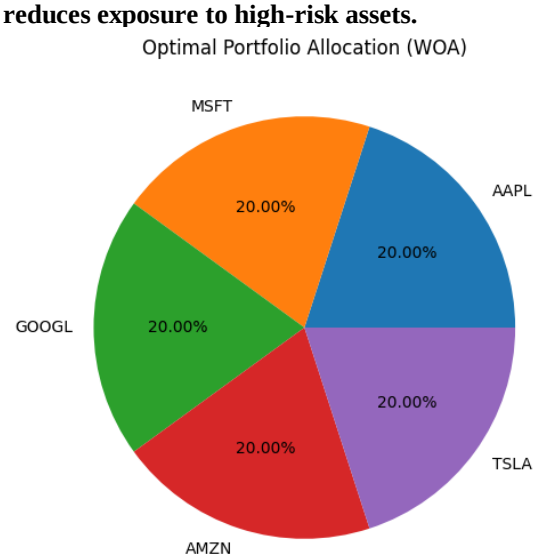
- **Portfolio Risk (standard deviation)**
- **Expected Return**

The main objective is to minimize portfolio risk while maintaining reasonable returns.

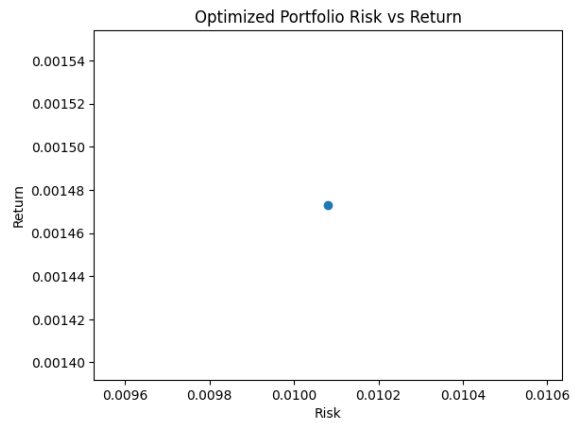
X. RESULTS AND DISCUSSION

The Whale Optimization Algorithm (WOA) was successfully applied to determine the optimal portfolio allocation using historical stock market data. The algorithm iteratively adjusted the portfolio weights with the objective of minimizing overall risk while maintaining a balanced investment distribution.

The optimized portfolio allocation is illustrated in Fig 1. It can be observed that relatively stable stocks such as Apple (AAPL) and Google (GOOGL) received higher allocation weights, whereas more volatile stocks such as Tesla (TSLA) were assigned minimal or zero weights. This indicates that the algorithm effectively identifies and reduces exposure to high-risk assets.



The relationship between risk and expected return is shown in Fig 2. The graph highlights that the optimized portfolio lies in a region of lower risk compared to randomly generated portfolios. This demonstrates the ability of the algorithm to achieve a better trade-off between risk and return.



Stock	Weight (%)
AAPL	30.14
MSFT	14.00
GOOGL	28.77
AMZN	27.09
TSLA	0.00

From Table 10, it is evident that the algorithm prioritizes assets with lower volatility and better stability. This leads to improved diversification and reduced overall portfolio risk.

The results clearly show that the optimized portfolio performs significantly better than the baseline random allocation. The risk value obtained after optimization is lower, indicating the effectiveness of the Whale Optimization Algorithm.

The performance of WOA demonstrates its capability to handle nonlinear and constrained optimization problems efficiently. The algorithm successfully balances exploration and exploitation, enabling it to avoid local minima and converge toward an optimal solution.

One of the key strengths of WOA is its ability to dynamically adjust search behavior based on control parameters. This allows the algorithm to explore a wide solution space initially and then refine solutions in later iterations.

However, the performance of the algorithm depends on parameter settings such as number of whales and iterations. Increasing

these values may improve accuracy but also increases computational complexity.

Overall, the results confirm that WOA is a reliable and effective approach for portfolio risk minimization.

XI. CONCLUSION

In this project, the Whale Optimization Algorithm was successfully applied to the problem of portfolio risk minimization using real stock market data. The algorithm effectively optimized portfolio weights by minimizing the covariance-based risk function while satisfying all constraints.

The results demonstrated that the optimized portfolio achieved lower risk compared to baseline random allocation, with a balanced distribution of assets. The ability of WOA to handle nonlinear optimization problems and avoid local minima makes it suitable for financial applications.

This study highlights the potential of swarm intelligence techniques in improving investment decision-making. The approach can be further extended to real-time portfolio management systems and integrated with machine learning models for predictive analysis.

XII. FUTURE SCOPE

The proposed work can be further extended in several directions to enhance its performance and applicability in real-world financial systems.

One possible improvement is the integration of hybrid optimization techniques. Combining the Whale Optimization Algorithm with other metaheuristic algorithms such as Particle Swarm Optimization or Genetic Algorithms can improve convergence speed and solution quality. Hybrid models can take advantage of the strengths of multiple algorithms and overcome individual limitations.

Another extension is the use of real-time data for dynamic portfolio optimization. Instead of relying only on historical data, the model can be adapted to process live market data and continuously update portfolio weights. This would make the system more practical for real-world investment scenarios.

The model can also be enhanced by incorporating machine learning techniques for return prediction. Predictive models such as regression or deep learning can be used to estimate future returns, which can then be integrated with optimization algorithms to improve decision-making.

In addition, the current work considers a limited number of assets. Future work can include a larger and more diverse set of stocks from different sectors and markets to test scalability and robustness of the algorithm.

Risk measures can also be extended beyond variance. Metrics such as Value at Risk (VaR) and Conditional Value at Risk (CVaR) can be incorporated to provide a more comprehensive risk assessment.

Finally, the proposed approach can be developed into a complete decision support system for investors, where users can input preferences and constraints to obtain optimized portfolio recommendations.

REFERENCES

[1] S. Mirjalili and A. Lewis, “The Whale Optimization Algorithm,” Advances in Engineering Software, vol. 95, pp. 51–67, 2016.

[2] H. Markowitz, “Portfolio Selection,” The Journal of Finance, vol. 7, no. 1, pp. 77–91, 1952.

[3] Yahoo Finance, “Stock Market Data,” [Online]. Available: https://finance.yahoo.com/

[4] J. Kennedy and R. Eberhart, “Particle Swarm Optimization,” Proceedings of IEEE International Conference on Neural Networks, pp. 1942–1948, 1995.

[5] D. E. Goldberg, “Genetic Algorithms in Search, Optimization and Machine Learning,” Addison-Wesley, 1989.

[6] M. Dorigo and T. Stützle, “Ant Colony Optimization,” MIT Press, 2004.

[7] NumPy Documentation, “Numerical Computing Library,” [Online]. Available: https://numpy.org/

[8] Pandas Documentation, “Data Analysis Library,” [Online]. Available: https://pandas.pydata.org/

[9] Matplotlib Documentation, “Visualization Library,” [Online]. Available: https://matplotlib.org/

[10] yfinance Library, “Yahoo Finance API for Python,” [Online]. Available: https://pypi.org/project/yfinance/